

Assignment 4

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Question 1

Reason 1: The Geocentric Model states that the earth is not moving, which is **consistent with our daily experience**. On the other hand, accepting the Heliocentric Model means that we need to accept a moving earth, which is against our understanding.

Reason 2: According to **Aristotelian's natural philosophy**, heavy objects will fall to the center of the universe. The earth is heavy, so it is in the center of the universe rather than orbiting the sun.

Reason 3: Traditional cosmology believes that imperfect and unpredictable objects lie on the earth, while perfect and predictable objects surround the celestial sphere. The earth is imperfect, so it is unreasonable that the earth orbits the sun.

Reason 4: Religious teachings hold that humanity stands at the center of the universe. Therefore, the heliocentric model contradicts religious beliefs.

Question 2

Aristotelian argument

If the earth is moving, we will land in a **different position** when jumping.

Tycho Brahe's argument

If the Earth moves, **parallax** should be observed when viewing the same star from different positions in Earth's orbit.

Galileo's moving boat experiment

Galileo's moving boat experiment refuted **Aristotle's argument**. When dropping an object on a moving boat, the object will land exactly at same position on the boat, rather than landing somewhere in the opposite moving direction. Therefore, even if the earth is moving, we will still land in the same place after jumping.

Question 3

We **cannot be certain** that the basic concepts that the models based on are true. Our current theories and models are based on our observation. It only provide us with a a model that work well in practice, but not an absolutely correct one. To disprove a model, we only need one counterexample; However, to prove a model, the model should work for all the objects. Generally speaking, the developing of science is through revising the old models and theorems by newly more accurate ones. Thus, we may never be able to tell whether some concepts are absolutely correct.